2	**(a)**			The hot gases exit (the rear of) the rocket with momentum The rocket must (momentarily) acquire an equal and opposite momentum resulting in the forward thrust.	B1 B1
	(b)			There is no drag force due to air resistance when the rocket is in vacuum When in air, drag force due to air resistance opposes the motion of the rocket, leading to a smaller resultant force.	B1 B1
	(c)	**(i)**		thrust (upwards, longer arrow) weight (downwards, shorter arrow)	B1 B1
		(ii)		$\text{thrust} - mg = ma$ $\text{thrust}\left(= ma + mg\right) = 1.2 \times 10^{6} \times \left(3.5 + 9.81\right)$ $= 1.60 \times 10^{7}$ N	 M1 B1
		(iii)		$\text{thrust} - mg = ma \Rightarrow \quad m(g+a) = \text{thrust} = 1.60 \times 10^{7}$ $m = \dfrac{1.60 \times 10^{7}}{9.81 + 18.7} = 561207$ kg $\dfrac{\Delta m}{\Delta t} = \dfrac{1.2 \times 10^{6} - 561207}{60} = 1.06 \times 10^{4}$ kg s^{-1}	 C1 A1
				[Total: 10]	